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CLEAVER BROOKS BOILER START UP PROCEDURE

This procedure details the steps required to safely bring the C.B. Boiler online. It also includes provisions to bring the CCM plant online from a cold state. This procedure assumes that the C.B. Boiler is cold, offline, and has insufficient water or is empty of water.

ASSUMPTIONS:

- Power and water systems are in service.
- Permission has been received for plant startup.

Boiler Walk Around and Auxiliary System Checks

Step	Operation	Initial when Complete	Date and Time when Complete
1	Check the Plant Log Book to make sure it is safe to start the C.B. Boiler. Inform Chief Engineer and Operations Centre of boiler start up.		
2	Perform a walk-around of the C.B. Boiler, checking fittings, valve positions, gauges and gauge cocks, interlock-and-permissive switches, force-draft-fan dampers, linkages and connections and the gas train.		
3	Check that the C.B. Boiler's feedwater pump is ready for service and associated valves are in the proper position. Check that there is ample feedwater available (refer to the Feedwater System Procedure).		
5	Check that the main stop valve is closed (the chain-operated valve).		

6	Open the bypass around the steam trap located downstream of the steam-outlet check valve. (This will reduce water hammer.)		

Special Circumstance - Plant Cold Startup

This step is to be done when the plant piping is cold. Performing this step will reduce steam and hydraulic water hammer. **If the plant piping is hot, skip to step 10.**

7	Check with the Plant Log Book to make sure there is no condition that will prevent the safe startup of the plant.		
8	Coordinate the start-up of the plant with the Operations Centre.		
9	Partially open the drip leg drains on the main steam header and the low pressure header. 45psi and 12psi header.		

Note: Use drain hoses to direct cold condensate to the sewer drain to prevent flooding the floors. Stop using drain hoses when the condensate becomes hot.

Fill the C.B. Boiler Steam Drum to the Proper Operating Level

Step	Operation	Initial when Complete	Date and Time when Complete
10	Open the steam drum vent valve if the boiler is being started from a cold, empty, or no pressure state.		
11	Turn on the boiler’s control power disconnect switches located on the post in front of the boiler and on the side of the boiler BMS control wiring box.		
12	The Weishaupt Display Screen and the Level Master PLCs will initialize. It the drum level is below normal, the Level Master PLC will start the boiler's feedwater pump and begin filling the boiler. If the drum level is too low, or too high an alarm will sound and continue to sound until the level in the drum rises to a safe level. If the level is too high then safely bottom blowdown boiler to achieve operating level.		

13	Reset the probe and auxiliary LWCOs as well as the boiler BMS system after the boiler is filled to the proper operating level.		
14	Leave the vent valve open and proceed with firing the boiler.		

Firing the Boiler

Step	Operation	Initial when Complete	Date and Time when Complete
Note: If the boiler is cold and not under pressure, ensure the steam drum vent valve is open.			
15	Place the Weishaupt HMI screen into firing rate controller into MANUAL and set the firing rate as low as possible (minimum fire, which is 10 %).		
16	Review the Weishaupt alarm page. Make sure there are no current alarms that would prevent the burner firing sequence.		
17	On the Weishaupt burner the Remote Control Screen place the toggle switch into “Hand” mode.		
18	Place the Weishaupt HMI on the Boiler Overview Screen.		
Note: If the Weishaupt HMI considers all permissives satisfied, it will initiate the startup sequence.			
19	Allow the boiler to warm up and gain pressure at minimum fire once the main flame is established.		
Note: The Weishaupt HMI will keep the boiler in a warm up mode until a preset drum pressure is reached. (on automatic mode only)			

Note: If the firing rate controller is in auto, the firing rate may spike to 100 percent once the warm up mode is automatically deactivated. To keep the boiler firing rate from "running away", make sure the boiler firing rate controller is in manual. The firing rate will be placed into automatic in step 34 of this procedure.

20	Close the steam drum vent when the boiler reaches 10 PSIg pressure.		
21	After reaching 15 to 20 PSIG, gradually increase the firing rate to about 10 percent.		
22	Slowly begin opening the main stop valve when the boiler reaches 50 PSIg.		

Bringing the Boiler Online

Step	Operation	Initial when Complete	Date and Time when Complete
23	Continue slowly opening the main stop valve and watch for condensate and then steam to flow from the free-blow floor drains on the headers (if starting the plant from a cold state).		
24	Close drip leg drains once all condensate is drained and pressure is sufficient to for the traps to function and blow condensate through to the hotwell.		

Note: Monitor the steam pressure ensuring it is increasing to the various steam users in the facility.

Note: This process of warming the boiler and steam systems to full pressure and temperature from cold is expected to take between half-an-hour in warm weather and up to two hours during cold winter weather.

Placing the Boiler into Automatic

25	Continue raising the boiler pressure by manually increasing the firing rate.		
26	Continue gradually opening the main stop valve until it’s fully open.		

<table>
<tr><td>27</td><td>Place the boiler’s firing rate controller into automatic once the boiler has come within 10 PSIG of its modulating set point and the drum level is stable.</td><td></td><td></td></tr>
<tr><td colspan="4">Set Point Settings</td></tr>
<tr><td>28</td><td colspan="3">Verify the following set points on the Clayton OIU on a weekly basis:
Modulating Set-Point (Weishaupt HMI): 95PSIg
Pressure Cut-Out-Point: 120 PSIg
Pressure Cut-In-Point: 105 PSIg
High-Pressure Trip: 135 PSIg
High-High-Pressure Trip: 140PSIg</td></tr>
<tr><td>29</td><td colspan="3">Make an appropriate log book entry.</td></tr>
<tr><td colspan="4">End of Procedure</td></tr>
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